PYTHON

INPUT AND OPERATORS PROBLEMS

1. Program for taking different inputs and performing operations

```
# Taking inputs of different data types

# Integer input

num1 = int(input("Enter an integer: "))

# Float input

num2 = float(input("Enter a float value: "))

# String input

text = input("Enter a string: ")

# Boolean input (convert from string)

bool_input = input("Enter True or False: ")

flag = bool_input.lower() == "true"

print("\n--- Input Values ---")

print("Integer:", num1)

print("Float:", num2)

print("String:", text)

print("Boolean:", flag)


# Performing operations

print("\n--- Operations ---")

# Arithmetic operations

print("Addition :", num1 + num2)

print("Subtraction:", num1 - num2)

print("Multiplication:", num1 * num2)

print("Division:", num1 / num2)

# String operations
```

```
print("String length:", len(text))

print("Uppercase:", text.upper())

print("Concatenation:", text + " - Python")

# Boolean operation

print("Boolean AND (flag and True):", flag and True)

# Type conversion examples (Explicit Typecasting)

print("\n--- Type Conversions ---")

print("Integer to float:", float(num1))

print("Float to int:", int(num2))

print("Integer to string:", str(num1))
```

OUTPUT:

```
Enter an integer: 34
Enter a float value: 9.3
Enter a string: HELLO
Enter True or False: true

--- Input Values ---
Integer: 34
Float: 9.3
String: HELLO
Boolean: True

--- Operations ---
Addition (int + float): 43.3
Subtraction: 24.7
Multiplication: 316.20000000000005
Division: 3.6559139784946235
String length: 5
Uppercase: HELLO
Concatenation: HELLO - Python
Boolean AND (flag and True): True

--- Type Conversions ---
Integer to float: 34.0
Float to int: 9
Integer to string: 34
```

2. Write a Python program to calculate the final price of a product.

- If the price is greater than 1000 and the user is a member, give a 20% discount.
- If the price is greater than 1000 but not a member, give a 10% discount.
- Otherwise, no discount is applied.
  Display the discount amount and final price.

```
price = float(input("Enter product price: "))
is_member = input("Are you a member (yes/no): ").lower() == "yes"

# Apply discount
if price > 1000 and is_member:
    discount = price * 0.2
elif price > 1000:
    discount = price * 0.1
else:
    discount = 0

final_price = price - discount

print("Discount:", discount)
print("Final Price:", final_price)
```

OUTPUT:

```
Enter product price: 1290
Are you a member (yes/no): yes
Discount: 258.0
Final Price: 1032.0
```

3. Write a program to accept marks of a student and assign grades based on the following:

- Marks ≥ 90 → Grade A
- Marks between 75 and 89 → Grade B
- Marks between 50 and 74 → Grade C
- Marks < 50 → Fail

```
marks = int(input("Enter marks: "))

if marks >= 90:
    grade = "A"
elif marks >= 75 and marks < 90:
    grade = "B"
elif marks >= 50:
    grade = "C"
else:
    grade = "Fail"

print("Grade:", grade)
```

```
Enter marks: 78
Grade: B
```

4. Create a simple login system that checks whether the entered username and password match the predefined credentials.

- Username: "admin"
- Password: "1234"
  Display "Login Successful" or "Invalid Credentials".

```
username = input("Enter username: ")
password = input("Enter password: ")

if username == "admin" and password == "1234":
    print("Login Successful")
else:
    print("Invalid Credentials")
```

OUTPUT:

```
Enter username: admin
Enter password: 1234
Login Successful
```

5. Write a program to check whether a given number is even or odd using operators.

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

OUTPUT:

```
Enter a number: 43
Odd Number
```

6. Develop a simple calculator program that takes two numbers as input and performs:

- Addition
- Subtraction
- Multiplication
- Division
- Modulus

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Modulus:", a % b)
```

OUTPUT:

```
Enter first number: 453
Enter second number: 35
Addition: 488
Subtraction: 418
Multiplication: 15855
Division: 12.942857142857143
Modulus: 33
```

7. In a system, permissions are represented using numbers:

- Read = 1
- Write = 2
- Execute = 4

Write a program to check which permissions a user has based on a given number using bitwise operators.

```
# Permissions: Read=1, Write=2, Execute=4
user_permission = 3  # Read + Write

if user_permission & 1:
    print("Read Permission Granted")
if user_permission & 2:
    print("Write Permission Granted")
if user_permission & 4:
    print("Execute Permission Granted")
```

OUTPUT:

```
Read Permission Granted
Write Permission Granted
```

8. Write a program to check whether a person is eligible to vote.
Conditions:

- Age must be 18 or above
- Person must be a citizen

```
age = int(input("Enter age: "))
citizen = input("Are you a citizen (yes/no): ").lower() == "yes"

if age >= 18 and citizen:
    print("Eligible to vote")
else:
    print("Not eligible")
```

OUTPUT:

```
Enter age: 23
Are you a citizen (yes/no): yes
Eligible to vote
```

9. Write a program to calculate tax based on salary:

- If salary > 500000 → 20% tax
- Otherwise → 10% tax
  Display tax amount and net salary.

```
salary = float(input("Enter salary: "))

if salary > 500000:
    tax = salary * 0.2
else:
    tax = salary * 0.1

print("Tax:", tax)
print("Net Salary:", salary - tax)
```

OUTPUT:

```
Enter salary: 15000
Tax: 1500.0
Net Salary: 13500.0
```

10. Write a program to validate an email address.
An email is considered valid if it contains both "@" and "." symbols.

```
email = input("Enter email: ")

if "@" in email and "." in email:
    print("Valid Email")
else:
    print("Invalid Email")
```

OUTPUT:

```
Enter email: 22IT932@GMAIL.COM
Valid Email
```

11. Write a program to swap two numbers without using a third variable.

```
a = int(input("Enter a: "))
b = int(input("Enter b: "))

a, b = b, a

print("After swapping:")
print("a =", a)
print("b =", b)
```

OUTPUT:

```
Enter a: 43
Enter b: 56
After swapping:
a = 56
b = 43
```